Preliminaries of Machine Learning

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1 Introduction

This is some notes about preliminaries of machine learning.

2 Probability

2.1 Basic Definitions

Definition 2.1.1 (Probability Space)

A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where

- Ω is a non-empty set called the *sample space*;
- $\mathcal{F}$ is a σ-algebra of subsets of the sample space Ω;
- $\mathbb{P} : \mathcal{F} \to [0, 1]$ is a probability measure.

2.2 Lists of Probability Distributions

2.2.1 Discrete Distributions

Name	Support	Parameters	PMF	Mean, Variance
Bernoulli	$\{0, 1\}$	$p \in \{0, 1\}$	$\mathrm{Ber}(x\|p) = p^x(1-p)^{1-x}$	$\mathbb{E}[X] = p, \mathbb{V}[X] = p(1-p)$
Binomial	$\{0, 1, 2, \ldots, N\}$	$N, p \in [0, 1]$	$\mathrm{Bin}(x\|N, p) = \binom{N}{x} p^x(1-p)^{N-x}$	$\mathbb{E}[X] = Np, \mathbb{V}[X] = Np(1-p)$
Categorical	$\{0, 1, 2, \ldots, N\}$	$\boldsymbol{\theta}$	$\mathrm{Cat}(x\|\boldsymbol{\theta}) = \prod_{k=1}^{K} \theta_k^{\mathbb{I}(x=k)}$	
Multinomial	$\boldsymbol{x}$ where $\sum_{k=1}^{K} x_k = N$	$N, \boldsymbol{\theta}$	$\mathrm{Mult}(\boldsymbol{x}\|\boldsymbol{\theta}) = \binom{N}{x_1, x_2, \ldots, x_K} \prod_{k=1}^{K} \theta_k^{x_k}$	
Poisson	$\{0, 1, 2, \ldots\}$	$\lambda > 0$	$\mathrm{Pois}(x\|\lambda) = e^{-\lambda} \dfrac{\lambda^x}{x!}$	$\mathbb{E}[X] = \lambda, \mathbb{V}[X] = \lambda$

2.2.2 Continuous Distributions

Name	Support	Parameters	PDF	Mean, Variance
Gaussian	$\mathbb{R}$	μ, σ	$\mathcal{N}(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$	$\mathbb{E}[X] = \mu, \mathbb{V}[X] = \sigma^2$
Student t-distribution	$\mathbb{R}$	μ, σ	$\mathcal{T}_\nu(x\|\mu, \sigma^2) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\pi\nu}\Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$	$\mathbb{E}[X] = 0$ for $\nu > 0$, $\mathbb{V}[X] = \frac{\nu}{\nu-2}$ for $\nu > 2$
Cauchy	$\mathbb{R}$	μ, λ	$\mathcal{C}(x\|\mu, \lambda) = \frac{1}{\lambda\pi(\frac{x-\mu}{\lambda})^2}$	undefined
Laplace	$\mathbb{R}$	μ, b	$\text{Laplace}(x\|\mu, b) = \frac{1}{2b} \exp\left(-\frac{\|x-\mu\|}{b}\right)$	$\mathbb{E}[X] = \mu, \mathbb{V}[X] = 2b^2$
Gamma	$\mathbb{R}^+$	$a, b > 0$	$\text{Ga}(x\|\text{shape} = a, \text{scale} = b) = \frac{b^a}{\Gamma(a)} x^{a-1} e^{-xb}$	$\mathbb{E}[X] = ab, \mathbb{V}[X] = ab^2$
Exponential	$\mathbb{R}^+$	λ	$\text{Expon}(x\|\lambda) = \lambda e^{-\lambda x}$	$\mathbb{E}[X] = \lambda, \mathbb{V}[X] = \lambda^2$
Chi-squared	$\mathbb{R}^+$	none	$\chi^2_\nu(x) = \text{Ga}(x\|\text{shape} = \nu/2, \text{scale} = 2)$	$\mathbb{E}[X] = \nu, \mathbb{V}[X] = 2\nu$

3 Statistics

This part is about guessing the parameters of our probability models shown in the previous section. Most of the estimation will take the form of $\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}}\ \mathcal{L}(\boldsymbol{\theta})$, where the $\boldsymbol{\theta}$ is our parameter to be estimated, and the $\mathcal{L}$ is the *loss function*.

There will be two main approaches, namely **frequentist statistics** and **Bayesian statistics**. Frequentist method treats parameters $\boldsymbol{\theta}$ as fixed value, and the data $\mathcal{D}$ as unknown and changes. Conversely, Bayesian method treats parameters $\boldsymbol{\theta}$ as unknown, and the data $\mathcal{D}$ as fixed and known.

However, note that the model may not take any parameters; such models can be adjusted by using **nonparametric methods**.

3.1 Bayesian Statistics

Bayesian statistics treats parameters $\boldsymbol{\theta}$ as unknown, and the data $\mathcal{D}$ as fixed and known. We use Bayes rule to update our uncertainty about parameters when the data is given;

$$p(\boldsymbol{\theta}|\mathcal{D}) = \frac{p(\boldsymbol{\theta})p(\mathcal{D}|\boldsymbol{\theta})}{p(\mathcal{D})} = \frac{p(\boldsymbol{\theta})p(\mathcal{D}|\boldsymbol{\theta})}{\int p(\boldsymbol{\theta}')p(\mathcal{D}|\boldsymbol{\theta}')d\boldsymbol{\theta}'}$$

Here, $p(\boldsymbol{\theta})$ is called the **prior**, and $p(\mathcal{D}|\boldsymbol{\theta})$ is called the **likelihood**. $p(\boldsymbol{\theta}|\mathcal{D})$ is called the **posterior**, and $p(\mathcal{D})$ is called the **marginal likelihood**.

After the calculation of posterior distribution, we can try to predict the future observations by calculating the **posterior predictive distribution**:

$$p(x|\mathcal{D}) = \int p(x|\boldsymbol{\theta})p(\boldsymbol{\theta}|\mathcal{D})d\boldsymbol{\theta}$$

We are marginalizing out all the unknown parameters. If the calculation is too expensive, we can just plug-in the paramter value we estimated, like MLE or MAP.

$$p(x|\mathcal{D}) \approx p(x|\hat{\boldsymbol{\theta}})$$

This is called **plug-in approximation**.

3.1.1 Conjugate Priors

Calculating the posterior distribution can be untractable, if the prior and likelihood does not match well. Here the well-matching prior according to the likelihood is given, called the **conjugate priors**.

likelihood	unknown parameters	conjugate prior	parameters(prior)	posterior

4 Decision Theory

5 Graphical Models

6 Information Theory

7 Optimization